Retrieval-Conditional Parsing Score (RCPS): Choosing Document Parsers by Retrieval, Not by Appearance

Sang-Woo Son, Hyeong-seob Kim, Hyeonsang Kim,
Hyun-woo Cho, Jinmo Kim

Abstract

Document RAG ultimately depends on retrieval, yet parsers are commonly selected by intrinsic measures of transcription fidelity or boundary quality. We introduce the Retrieval-Conditional Parsing Score (RCPS), a training-free protocol that ranks parser–chunker combinations on a fixed, held-out question–answer (Q–A) retrieval probe. RCPS averages standard MRR across specified retrievers and cutoffs under one format-normalised exact-span relevance rule. On KoGovDoc-RAG, table-enabled MinerU-on has higher Boundary Clarity than our production parser, Prod (0.713 versus 0.610), but much lower Hit@1 (0.123 versus 0.549), a 42.6-percentage-point gap. Among five parser configurations with complete 294-page outputs, RCPS spans 0.137–0.584; with Prod fixed, four chunkers span 0.535–0.593. A complementary retriever-free diagnostic finds that 20.2% of reference spans have no normalised exact match in Prod output, whereas at most 2.3% are split by chunk boundaries. On source-aligned OHR-Bench data, semantic noise reduces retrieval while Boundary Clarity remains stable or changes non-monotonically. In a secondary parser-training study, two direct preference optimization checkpoints have Hit@5 estimates 0.95 and 1.15 percentage points above Prod on an audited six-domain compatibility subset. The resulting workflow is to select with RCPS, diagnose with coverage, and train only if needed. We release the 663-Q–A KoGovDoc-RAG probe and an RCPS reference implementation; full source-page mapping and some rerun artifacts remain pending.

1 Introduction

Document RAG depends on parser output, yet parsers are commonly selected by transcription fidelity or Boundary Clarity (BC) (Zhao et al., 2025a) rather than by retrieval. Because parsing precedes chunking, embedding, and retrieval, omitted or corrupted content can cap downstream performance. Intrinsic metrics describe the output; they do not directly measure whether it preserves retrievable answers.

We therefore organise deployment into three steps: select by retrieval with RCPS, diagnose parser- versus chunker-side loss with coverage, and consider parser training only afterward (Figure 1).

Our production results show why selection must follow retrieval. Table-enabled MinerU-on (Niu et al., 2025) has higher BC than Prod (0.713 versus 0.610) but reaches only 0.123 Hit@1 versus Prod's 0.549, a 42.6-point gap. On source-aligned OHR-Bench data (Zhang et al., 2025a), semantic corruption likewise lowers retrieval while BC remains stable or non-monotonic.

Coverage then locates the loss: 20.2% of reference spans have no normalised exact match in Prod output before chunking, while no tested chunker splits more than 2.3%.

Training is secondary. After selection and diagnosis, we separately test whether parser-side training with retrieval-oriented feedback improves Prod. Two direct preference-optimization checkpoints are only about one Hit@5 point above Prod on an audited OHR-Bench compatibility subset. This frame differs from the selection comparison, so the effect sizes are not directly comparable.

Contributions. We show that intrinsic metrics can misrank parsers (C1); define RCPS for selecting parsers and chunkers (C2); separate parser-side absence from chunker-side splitting (C3); and quantify the smaller,

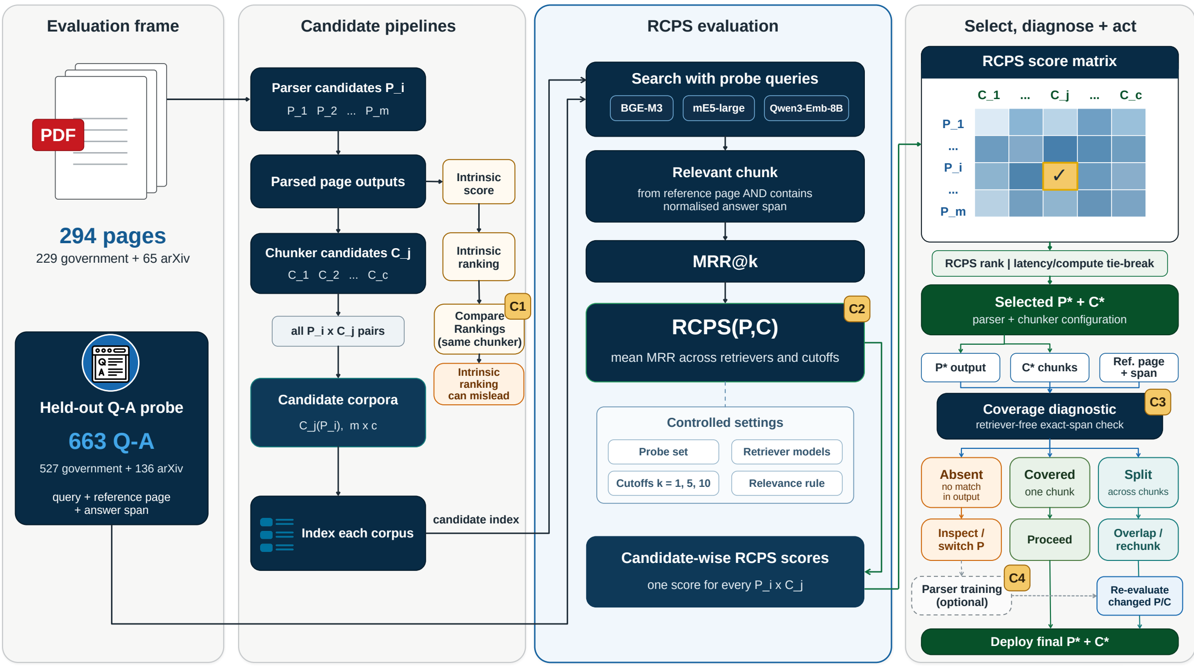

Figure 1: **RCPS workflow.** A held-out Q–A probe evaluates parser–chunker candidates. Intrinsic and RCPS parser rankings can disagree (C1); RCPS ranks candidates across fixed retrievers and cutoffs (C2); coverage separates spans absent from parser output from those split across chunk boundaries (C3); and parser training is reserved for unresolved parser-side failures before changed configurations are re-evaluated (C4).

uncertain differences associated with parser training (C4). Our scope is answer-span retrievability. Appendix B checks end-to-end answering for three parsers.

The repository contains the frozen 663-Q–A **KoGovDoc-RAG** probe and a reference implementation of RCPS.[1] The selection experiment indexed 294 pages, but the portable source-page mapping and some rerun artifacts are not yet packaged (Appendix H).

2 Related Work

Evidence and gap. Existing benchmarks establish the parsing–retrieval mismatch; we turn it into a selection-and-diagnosis workflow. OHR-Bench (Zhang et al., 2025a) traces OCR noise through RAG, while Enterprise-DocBench (Singh and Raj, 2026) and concurrent studies (Sun et al., 2026; Song, 2026) find weak associations between parsing quality and retrieval. Most parsing benchmarks still rank intrinsic fidelity (Ouyang et al., 2025). Vision-based retrieval (Faysse et al., 2025) instead bypasses text parsing. We are unaware of prior work that combines a reusable parser-selection protocol with a diagnostic that separates missing from boundary-split spans.

[1] https://github.com/wigtn/WigtnOCR-RADP

Training. Retrieval-oriented training usually targets chunkers (Günther et al., 2024; Duarte et al., 2024; Zhao et al., 2024, 2025a; Singh et al., 2025), embeddings (Conti et al., 2025; Zhao et al., 2025b), retrievers, or readers (Nguyen et al., 2024; Chia et al., 2025; Yan et al., 2025); our secondary study intervenes at the parser. Prior parser training uses edit-distance and layout rewards (Wang et al., 2025) or query guidance (Wang et al., 2026). We test a retrieval reward and design a fidelity control, but its aligned per-Q–A results are unavailable, so C4 makes no matched objective comparison. C1–C3 require no training.

3 Two Tools: Select and Diagnose

RCPS and coverage answer different deployment questions. RCPS asks which parser–chunker combination retrieves best; coverage asks whether an exact span is absent from parser output or divided during chunking.

3.1 RCPS Selects by Retrieval

RCPS compares candidates on the same held-out Q–A probe. It wraps standard truncated mean reciprocal rank (MRR), rather than defining a new similarity function. Three choices keep comparisons consistent: evaluation runs on each parsed-and-chunked corpus; scores average specified retrievers; and every candidate uses the same *format-normalised exact-span relevance*. A chunk is relevant only if it comes from the reference page and contains the reference answer after shared whitespace and Markdown normalisation. We call Equation 1 the RCPS *score* and the complete procedure the RCPS *protocol*.

Given a parser P, a chunker C, a Q–A set $D = \{(q_i, a_i, \text{page}_i)\}$, a set of retrievers R, and cutoffs K, RCPS averages MRR over all retriever–cutoff pairs:

$$\text{RCPS}(P, C) = \frac{1}{|R|\,|K|} \sum_{r \in R} \sum_{k \in K} \text{MRR@}k(r, C(P), D). \tag{1}$$

Here, $C(P)$ denotes the corpus obtained by parsing with P and then chunking with C. We instantiate R with BGE-M3 (Chen et al., 2024), multilingual-e5-large (Wang et al., 2024), and Qwen3-Embedding-8B (Zhang et al., 2025b), and use $K = \{1, 5, 10\}$. Averaging over cutoffs avoids tying selection to one retrieval depth. Figure 2 shows the workflow.

Execution. Build a representative held-out probe whose answers occur verbatim in their reference pages. Our probe is model-generated; an LLM-based check accepted 94/100 sampled pairs (§4.1). Parse each page once per parser, then chunk, index, and search every candidate corpus. For m parsers, c chunkers, and $|R|$ retrievers, this requires m parsing runs and $mc|R|$ retrieval evaluations.

Interpretation. The protocol needs neither training nor manual chunk labels because relevance follows from the answer span and source page. Fixing C ranks parsers; fixing P ranks chunkers. The ranking is relative to the candidate pool and probe.

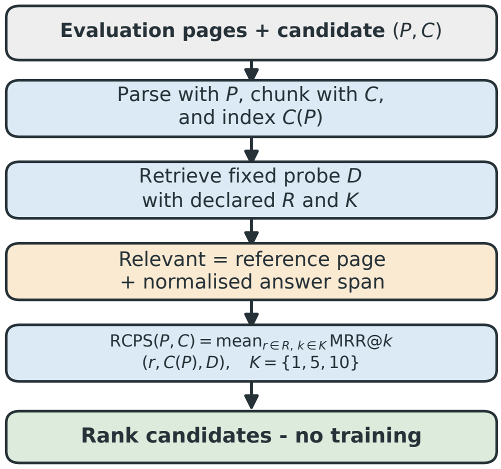

Figure 2: RCPS compares candidates on the same held-out Q–A probe. It averages standard MRR over declared retrievers and cutoffs; a hit must come from the reference page and contain the normalised reference answer. Use a singleton retriever set when deployment is fixed, or a declared set when it is undecided. Evaluation itself requires no training.

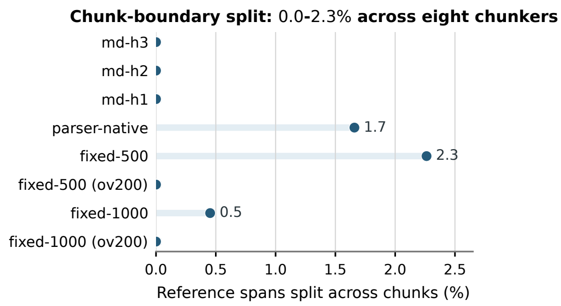

Figure 3: Coverage with Prod fixed (294 pages, 663 Q–A; no retriever). Absence is measured before chunking and stays 20.2%; changing the chunker affects only splitting, which reaches at most 2.3%.

3.2 Coverage Separates Absence from Splitting

Coverage separates spans absent from parser output from spans split across chunk boundaries; it does not score retrievers. With parser output fixed, an answer is *covered* if its normalised span appears within one chunk, *split* if it exists on the page but crosses chunks, and *absent* if the page output contains no exact match. Overlap may repair a split, whereas rechunking alone cannot restore an absent exact span. These labels are operational: *absent* does not prove semantic omission.

4 Experiments

4.1 Evaluation Frames and Setup

Evaluation frames. We report five non-interchangeable frames: three KoGovDoc-RAG frames of 294, 242, and 73 pages, and two OHR-Bench frames of 1,043 and 2,036 Q–A. The selection analysis retrieves all 663 KoGovDoc-RAG Q–A against 294 pages (229 government and 65 arXiv); their evidence occurs on 242 pages. Training and mechanism analyses index only those 242 pages, while the pilot uses 73 pages and 202 Q–A. OHR-Bench (Zhang et al., 2025a) uses 1,043 aligned Law–Manual Q–A for perturbations and a separate 2,036-Q–A, six-domain compatibility subset for training evaluation (Appendix A).

Data and leakage control. Qwen3-VL-30B produced pseudo-reference Markdown, which we manually de-noised; `gpt-5.4-2026-03-05` then generated the Q–A. An LLM-based check accepted 94/100 sampled pairs. **Prod**, a Qwen3-VL-2B parser (Bai et al., 2025) fine-tuned for Korean documents, is the training baseline.

For MinerU, *MinerU-off* denotes the submitted output without table recognition, while *MinerU-on* denotes the later audited, table-enabled output. The deployment comparison and its four-parser BC correlation use MinerU-on; MinerU-off is retained only as a separately labelled submitted-output diagnostic. Because the two runs also differ in software and retrieval environments, their differences do not isolate table recognition.

4.2 Boundary Clarity Can Misrank Parsers in This Pool (C1)

Boundary Clarity does not track RCPS in our candidate pool: outputs with the clearest boundaries can rank near the bottom on retrieval. Across five complete 294-page deployment configurations, RCPS spans 0.137–0.584; vision–language parsers rank above OCR systems (Figure 4; Table 1). The 30B teacher and Prod lead at 0.584 and 0.583. MinerU-on has the highest measured BC among the complete-output parsers (0.713) but the lowest RCPS (0.137); PaddleOCR lacks measurable BC. Marker's 0.073 score covers only 38 pages and is excluded from complete-output comparisons.

Among four 294-page parsers with defined BC, the Pearson correlation between BC and RCPS is $r = -0.74$. Adding Marker's 38-page result gives $r = -0.83$ ($n = 5$). MinerU-on and Marker have the highest BC values (≈ 0.71) but the lowest retrieval ranks. Both small-sample correlations are descriptive.

Law–Manual perturbations (1,043 Q–A). Within this aligned subset, semantic noise always lowers RCPS, whereas BC is stable, non-monotonic, or increasing. The study contains three benchmark outputs and twelve perturbations, not fifteen independent parsers (Appendix A). From mild to severe noise, MinerU RCPS falls $0.476 \rightarrow 0.265$ while BC varies within 0.628–0.651; Qwen2.5-VL RCPS falls $0.534 \rightarrow 0.497$ while BC stays near 0.563; and GOT RCPS falls $0.461 \rightarrow 0.298$ as BC rises $0.586 \rightarrow 0.624$. We claim no broader domain generality.

4.3 RCPS **Ranks Parsers and Chunkers (C2)**

On the same 663-Q–A probe and 294-page index, the 30B teacher and Prod are descriptively near-tied (0.584 versus 0.583); with Prod fixed, four chunkers form a point-estimate order from 0.593 to 0.535.

Selecting parsers. In Table 1, Prod reaches 0.583 RCPS and 0.549 Hit@1, versus MinerU-

Parser	BC	CS	RCPS	Hit@1
Deployment comparison (294 pages)				
Qwen3-VL-30B (teacher)	0.623	3.38	**0.584**	0.545
Prod (ours, 2B)	0.610	3.07	0.583	0.549
Qwen3-VL-2B (base)	0.520	3.74	0.532	0.500
PaddleOCR	—	3.46	0.140	0.125
MinerU-on	**0.713**	—	0.137	0.123
Submitted-output diagnostic (294 pages)				
MinerU-off	0.716	2.81	0.212	0.197
Partial output				
Marker (38p)	0.717	3.41	0.073	0.068

Table 1: RCPS ranks the 30B teacher and Prod at the top of the complete-output parser pool. Scores average three retrievers and $k \in \{1, 5, 10\}$. BC is Boundary Clarity (higher is better); CS is Chunk Stickiness (lower is better). The four-parser deployment correlation uses MinerU-on. MinerU-off is retained as a separate submitted-output diagnostic, and Marker is excluded from complete-output comparisons.

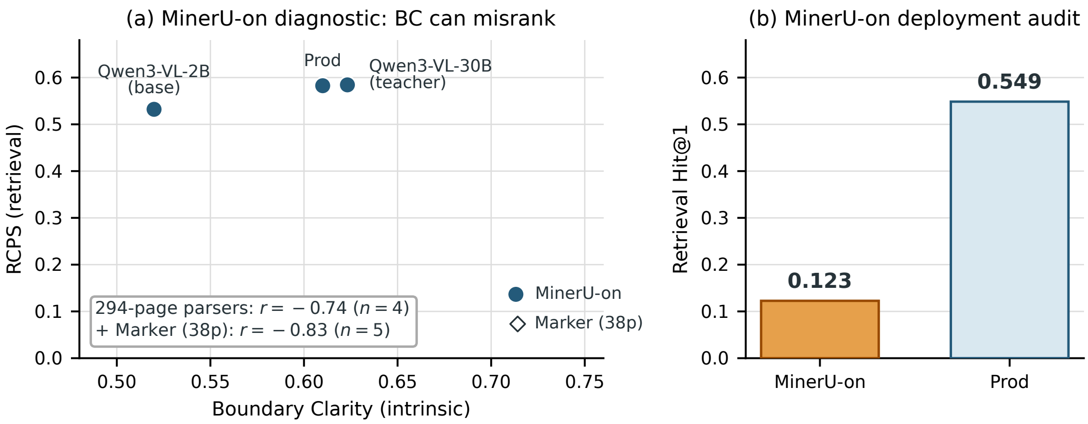

Figure 4: In this candidate pool, higher Boundary Clarity (BC) does not imply better retrieval. (a) Among four 294-page parsers with defined BC, the correlation with RCPS is $r = -0.74$; adding Marker's partial 38-page output gives $r = -0.83$ ($n = 5$). MinerU-on is used in this diagnostic. (b) Prod exceeds MinerU-on Hit@1 by 42.6 points (0.549 vs. 0.123; 4.47×).

on's 0.137 and 0.123; Prod's Hit@1 is 4.47× as high. The 30B teacher leads Prod by only 0.001 RCPS. Without paired uncertainty for this grid, we call them descriptively near-tied, not equivalent; latency and compute can break the deployment tie.

Selecting chunkers. With Prod fixed, the order is md-h3 (0.593) > parser-native (0.583) > LumberChunker (Duarte et al., 2024) (0.557) > fixed-500 (0.535). The chunker range is 0.058, versus 0.447 across the heterogeneous parser pool but 0.052 among only its three vision–language parsers.

Why average retrievers? Within the stored MinerU-off grid, BGE-M3 alone changes only the near-tied 30B–Prod order ($\tau = 0.80$; Table 2) and leaves the chunker order unchanged. This ablation predates the MinerU-on audit and is not used to compare MinerU-on with PaddleOCR. Use the deployment retriever when fixed; averaging hedges when it is undecided or candidates tie. An aggregate audit also finds that three-retriever MRR@10 preserves the stored five-parser and four-chunker orders ($\tau = 1.0$). Missing aligned per-Q–A arrays prevent a probe-resampling test.

Protocol configuration	Top pair	τ vs RCPS
BGE-M3 only	Prod ≻ 30B	0.80
full RCPS	30B ≻ Prod	—

Table 2: In the stored MinerU-off grid, three-retriever averaging changes only the near-tied 30B–Prod order; lower ranks are unchanged. The five complete-output configurations exclude Marker and keep relevance fixed. This ablation is separate from the MinerU-on deployment comparison.

4.4 Coverage Finds Pre-Chunking Absence (C3)

For Prod, 134/663 reference spans (20.2%) have no normalised exact match before chunking; no tested chunker splits more than 15/663 (2.3%) (Figure 3). This retriever-free diagnostic uses all 294 pages. Rechunking therefore cannot restore the exact match for roughly one answer in five.

MinerU-on's absent rate is 66.1%, a 45.9-point gap over Prod under the same normalised exact matcher. In the separately retained submitted-output analysis, MinerU-off's absent rate is 70.4%; its gaps over Prod remain 50.2 points under exact matching, 51.7 under an OCR-tolerant matcher, and 50.4 under GPT-5.4 retrieval-usability labels (Appendix C).

4.5 Parser Training Is a Secondary, Bounded Lever (C4)

We call the optional parser-training stage after RCPS selection and coverage diagnosis Retrieval-Aware Document Parsing (RADP). RADP-aux aligns answer-span hidden states with frozen BGE-M3 embeddings through an auxiliary contrastive loss. RADP-DPO samples multiple parses per page, ranks them with page-local BGE-M3 MRR, and converts higher- and lower-scoring parses into DPO preference pairs (Rafailov et al., 2023). RADP-Distill is a non-retrieval fidelity control that replaces the retrieval reward with edit distance to reference Markdown. Thus RCPS selects a pipeline without training, whereas RADP changes the parser after diagnosis.

All training data are page-disjoint from evaluation; Appendix E gives the common setup and preference construction. On the held-out 73-page pilot, neither RADP-aux nor RADP-DPO meets the target of a 5-point RCPS gain with a 95% CI lower bound above zero. On the audited 2,036-Q–A OHR-Bench compatibility subset, R2 and R3 exceed Prod Hit@5 by only 0.95 and 1.15 points (95% CIs $[+0.33, +1.54]$ and $[+0.31, +2.05]$). These post-audit estimates are neither a full v2 rerun nor a prespecified confirmation. Appendix F reports the full results, the reference-free SimPO control (Meng et al., 2024), and the unavailable aligned RADP-Distill comparison.

5 Discussion and Conclusion

Select, diagnose, then train. Parser choice changes Hit@1 by 42.6 points in the audited Prod–MinerU-on comparison; RADP-DPO is only about one Hit@5 point above Prod on a different OHR-Bench frame. These magnitudes are not directly comparable, but they make selection the first decision. In a three-parser end-to-end check, the RCPS top choice, Prod, also leads answer accuracy (72.5% versus 23.8% and 20.5%). The lower pair reverses, so this supports only the top choice.

Clear boundaries do not guarantee retrieval. BC measures boundary discontinuity, TextNED measures transcription similarity, and coverage tests exact-span preservation. MinerU-on combines high BC (0.713) with 66.1% exact-span absence and low Hit@1 (0.123), showing why these signals differ. DPO checkpoints have lower TextNED than Prod, but incomplete boundary evidence prevents a mechanism claim. Lower TextNED is therefore a post hoc association, not evidence that better fidelity caused retrieval gains (Appendix G).

Practical recommendation. Rank candidates with RCPS; add overlap for *split* spans, inspect *absent* spans, and switch parsers when required content is missing. Train only afterward. Neither RADP method met the pilot target, and SimPO had negative point estimates. Retrieval should therefore select the parser; coverage and fidelity diagnostics should guide what to improve next.

Limitations

Data scope and frames. Denominators are not interchangeable. KoGovDoc-RAG selection indexes 294 pages for 663 Q–A; training indexes the 242 evidence-bearing pages for the same Q–A; and the pilot uses 73 pages and 202 Q–A. The complete comparison covers five 294-page parsers. The descriptive BC correlations use the four with measurable BC ($r = -0.74$) or add Marker on 38 pages ($r = -0.83$). OHR-Bench uses 1,043 Law–Manual Q–A for perturbations and 2,036 Q–A for training compatibility. The former has three parser outputs plus twelve dependent perturbations and does not establish domain generality. We exclude mixed-version seven-domain artifacts (Appendix A).

MinerU audit. MinerU-off disables table recognition and is retained only as a submitted-output diagnostic. The separate MinerU-on audit reduces table-evidence absence from 87.9% to 41.7% (Prod: 13.9%), yet its mixed-corpus RCPS is 0.137 and Hit@1 is 0.123 (Prod: 0.583 and 0.549). MinerU-on and MinerU-off have numerically similar BC (0.713 and 0.716), but their software and retrieval environments also differ, so changes in BC, retrieval, and absence do not isolate table recognition. MinerU-on RCPS is 0.046 on government pages and 0.486 on arXiv pages.

Reference, Q–A, and label quality. Two 100-case checks answer different questions:

an LLM check accepts 94/100 sampled Q–A, whereas a blind two-author study validates coverage labels. KoGovDoc-RAG uses manually de-noised Qwen3-VL-30B pseudo-references and LLM-generated Q–A; neither complete set was human-verified. The fixed probe controls query variation, not possible Qwen-lineage bias. OHR-Bench provides external checks, but the six-domain result is a compatibility analysis, not a full v2 rerun. RCPS covers verbatim spans, not implicit or paraphrased answers. Coverage is matcher-defined: GPT-5.4 reclassifies 56% of Prod's exact-match-absent cases as surface artefacts. The same GPT-5.4 checkpoint generates and judges end-to-end answers, so that check supports only the top parser choice.

Training. KoGovDoc-RAG R2 is exploratory: its +1.96-point Hit@5 estimate (95% CI $[-1.06, +5.03]$) uses 242 pages and a configuration selected after inspecting several settings. On the separate 2,036-Q–A OHR compatibility subset, R2 and R3 are +0.95 $[+0.33, +1.54]$ and +1.15 $[+0.31, +2.05]$ points above Prod. This subset was defined after a version audit and is not a full v2 rerun. We neither adjust for multiple checkpoints and secondary metrics nor cluster bootstrap resamples by shared evidence page. These are post-audit estimates, not the original confirmatory analysis.

Scope. RCPS targets answer-span retrieval, not generated answers; the end-to-end experiment checks only three parsers. Preference candidates come from Prod, so other pools may differ. We do not test token-level reinforcement learning, curricula, or chunk-level oracle distillation. Broader parsers and training of chunkers, embedders, and rerankers remain future work.

References

Shuai Bai, Yuxuan Cai, Ruizhe Chen, Keqin Chen, Xionghui Chen, Zesen Cheng, Lianghao Deng, Wei Ding, Chang Gao, Chunjiang Ge, Wenbin Ge, Zhifang Guo, Qidong Huang, Jie Huang, Fei Huang, Binyuan Hui, Shutong Jiang, Zhaohai Li, Mingsheng Li, and 45 others. 2025. Qwen3-vl technical report. ArXiv:2511.21631.

Jianlv Chen, Shitao Xiao, Peitian Zhang, Kun Luo, Defu Lian, and Zheng Liu. 2024. M3-embedding: Multi-linguality, multi-functionality, multi-granularity text embeddings through self-knowledge distillation. Released as BGE-M3; arXiv:2402.03216.

Yew Ken Chia, Liying Cheng, Hou Pong Chan, Chaoqun Liu, Maojia Song, Sharifah Mahani Aljunied, Soujanya Poria, and Lidong Bing. 2025. M-longdoc: A benchmark for multi-modal super-long document understanding and a retrieval-aware tuning framework. In *Proceedings of EMNLP*. ArXiv:2411.06176.

Max Conti, Manuel Faysse, Gautier Viaud, Antoine Bosselut, Céline Hudelot, and Pierre Colombo. 2025. Context is gold to find the gold passage: Evaluating and training contextual document embeddings. InSeNT; arXiv:2505.24782.

André V. Duarte, João Marques, Miguel Graça, Miguel Freire, Lei Li, and Arlindo L. Oliveira. 2024. Lumberchunker: Long-form narrative document segmentation. In *Findings of EMNLP*. ArXiv:2406.17526.

Manuel Faysse, Hugues Sibille, Tony Wu, Bilel Omrani, Gautier Viaud, Céline Hudelot, and Pierre Colombo. 2025. Colpali: Efficient document retrieval with vision language models. In *International Conference on Learning Representations (ICLR)*. ArXiv:2407.01449.

Michael Günther, Isabelle Mohr, Daniel James Williams, Bo Wang, and Han Xiao. 2024. Late chunking: Contextual chunk embeddings using long-context embedding models. Jina AI; arXiv:2409.04701.

Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. 2022. Lora: Low-rank adaptation of large language models. In *International Conference on Learning Representations (ICLR)*. ArXiv:2106.09685.

Yu Meng, Mengzhou Xia, and Danqi Chen. 2024. Simpo: Simple preference optimization with a reference-free reward. In *Advances in Neural Information Processing Systems (NeurIPS)*. ArXiv:2405.14734.

Thang Nguyen, Peter Chin, and Yu-Wing Tai. 2024. Reward-rag: Enhancing rag with reward driven supervision. ArXiv:2410.03780.

Junbo Niu, Zheng Liu, Zhuangcheng Gu, Bin Wang, Linke Ouyang, Zhiyuan Zhao, Tao Chu, Tianyao He, Fan Wu, Qintong Zhang, Zhenjiang Jin, Guang Liang, Rui Zhang, Wenzheng Zhang, Yuan Qu, Zhifei Ren, Yuefeng Sun, Yuanhong Zheng, Dongsheng Ma, and 42 others. 2025. Mineru2.5: A decoupled vision-language model for efficient high-resolution document parsing. ArXiv:2509.22186.

Linke Ouyang, Yuan Qu, Hongbin Zhou, Jiawei Zhu, Rui Zhang, Qunshu Lin, Bin Wang, Zhiyuan Zhao, Man Jiang, Xiaomeng Zhao, Jin Shi, Fan Wu, Pei Chu, Minghao Liu, Zhenxiang Li, Chao Xu, Bo Zhang, Botian Shi, Zhongying Tu, and Conghui He. 2025. Omnidocbench: Benchmarking diverse pdf document parsing with comprehensive annotations. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. ArXiv:2412.07626.

Rafael Rafailov, Archit Sharma, Eric Mitchell, Stefano Ermon, Christopher D. Manning, and Chelsea Finn. 2023. Direct preference optimization: Your language model is secretly a reward model. In *Advances in Neural Information Processing Systems (NeurIPS)*. ArXiv:2305.18290.

Ishneet Sukhvinder Singh, Ritvik Aggarwal, Ibrahim Allahverdiyev, Muhammad Taha, Aslihan Akalin, Kevin Zhu, and Sean O'Brien. 2025. Chunkrag: Novel llm-chunk filtering method for rag systems. In *Proceedings of NAACL Student Research Workshop*. ArXiv:2410.19572.

Saurabh K. Singh and Sachin Raj. 2026. Benchmarking complex multimodal document processing pipelines: A unified evaluation framework for enterprise ai. ArXiv:2604.26382.

Minchae Song. 2026. Defining the problem: The impact of ocr quality on retrieval-augmented generation performance and strategies for improvement. *Information Processing & Management*, 63(1). Doi:10.1016/j.ipm.2025.104368.

Lin Sun, Wangdexian, Jingang Huang, Linglin Zhang, Change Jia, Zhengwei Cheng, and Xiangzheng Zhang. 2026. When good ocr is not enough: Benchmarking ocr robustness for retrieval-augmented generation. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 6: Industry Track)*, pages 884–894. ArXiv:2605.00911; doi:10.18653/v1/2026.acl-industry.60.

Baode Wang, Biao Wu, Weizhen Li, Meng Fang, Zuming Huang, Jun Huang, Haozhe Wang, Yanjie Liang, Ling Chen, Wei Chu, and Yuan Qi. 2025. Infinity parser: Layout aware reinforcement learning for scanned document parsing. ArXiv:2506.03197.

Liang Wang, Nan Yang, Xiaolong Huang, Linjun Yang, Rangan Majumder, and Furu Wei. 2024. Multilingual e5 text embeddings: A technical report. Microsoft; arXiv:2402.05672.

Zhengren Wang, Dongsheng Ma, Huaping Zhong, Jiayu Li, Wentao Zhang, Bin Wang, and Conghui He. 2026. Agenticocr: Parsing only what you need for efficient retrieval-augmented generation. ArXiv:2602.24134.

Shi-Qi Yan, Quan Liu, and Zhen-Hua Ling. 2025. Rpo: Retrieval preference optimization for robust retrieval-augmented generation. ArXiv:2501.13726.

Junyuan Zhang, Qintong Zhang, Bin Wang, Linke Ouyang, Zichen Wen, Ying Li, Ka-Ho Chow, Conghui He, and Wentao Zhang. 2025a. Ocr hinders rag: Evaluating the cascading impact of ocr on retrieval-augmented generation. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*. ArXiv:2412.02592.

Yanzhao Zhang, Mingxin Li, Dingkun Long, Xin Zhang, Huan Lin, Baosong Yang, Pengjun Xie, An Yang, Dayiheng Liu, Junyang Lin, Fei Huang, and Jingren Zhou. 2025b. Qwen3 embedding: Advancing text embedding and reranking through foundation models. ArXiv:2506.05176.

Jihao Zhao, Zhiyuan Ji, Zhaoxin Fan, Hanyu Wang, Simin Niu, Bo Tang, Feiyu Xiong, and Zhiyu Li. 2025a. Moc: Mixtures of text chunking learners for retrieval-augmented generation system. ArXiv:2503.09600.

Jihao Zhao, Zhiyuan Ji, Yuchen Feng, Pengnian Qi, Simin Niu, Bo Tang, Feiyu Xiong, and Zhiyu Li. 2024. Meta-chunking: Learning text segmentation and semantic completion via logical perception. ArXiv:2410.12788.

Yao Zhao, Yantian Ding, Zhiyue Zhang, Dapeng Yao, and Yanxun Xu. 2025b. Lmar: Language model augmented retriever for domain-specific knowledge indexing. ArXiv:2508.05672.

Appendix roadmap. Appendices A–D support the intrinsic-metric, RCPS-selection, and coverage analyses (C1–C3). Appendices E–G document the secondary parser-training study (C4), and Appendix H records dataset and release scope.

A OHR-Bench Alignment and Noise Sensitivity

We separate two valid OHR-Bench frames after finding that the initial seven-domain artifacts mixed benchmark versions. Those artifacts paired Q–A from the legacy 4,330-page release with parser outputs from the current 8,561-page bundle. For 223 legacy Q–A labelled `notes`, the old evaluation mapped `notes` to v2 `administration`, although the corresponding evidence resides under v2 `textbook`. The lookup therefore selected unrelated documents and stored zeros for all 223 cases. One additional legacy textbook page, referenced by five Q–A, is absent from v2. We exclude these 228 cases and retain a 2,036-Q–A, source-aligned compatibility subset across six domains. This is not a full v2 rerun, and we discard the mixed-version correlation.

The Law–Manual files retain matching document and page identifiers across the two releases. We therefore restrict the C1 perturbation study to that 1,043-Q–A subset. The 15 variants comprise three benchmark outputs (the reference output, MinerU, and Qwen2.5-VL), three formatting perturbations, and nine semantic perturbations spanning three severities for MinerU, Qwen2.5-VL, and GOT. No unperturbed GOT output is available. Across all variants, the descriptive BC–RCPS correlation is $r = -0.35$; variants within one family are not independent parsers. Table 3 reports the semantic families, using the same Law–Manual pages and three-retriever average throughout.

Family	Metric	Clean	Mild	Mod.	Severe
MinerU	BC	0.657	0.628	0.651	0.631
	RCPS	0.595	0.476	0.384	0.265
Qwen2.5-VL	BC	0.563	0.563	0.563	0.564
	RCPS	0.545	0.534	0.523	0.497
GOT	BC	—	0.586	0.620	0.624
	RCPS	—	0.461	0.385	0.298

Table 3: Semantic noise lowers RCPS within every family, while BC is stable, non-monotonic, or increasing. Results use 1,043 source-aligned Law–Manual Q–A. Clean denotes unperturbed output; Mild, Mod., and Severe denote increasing semantic noise. GOT has no clean output.

B Does the RCPS Top Choice Lead on Answering?

We test whether the RCPS top choice among Prod, MinerU-on, and PaddleOCR also achieves the highest answer-generation accuracy on all 663 KoGovDoc-RAG Q–A. For answer generation, BGE-M3 retrieves the top five chunks from each parser corpus; `gpt-5.4-2026-03-05` then generates an answer, and the same model checkpoint judges it against the reference answer. The displayed RCPS values follow the main protocol and average three retrievers at three cutoffs; they are not BGE-M3 top-five scores alone. For the MinerU row, both RCPS and answer generation use MinerU-on; the other rows use their corresponding parser outputs. The submitted MinerU-off output scores 0.212 RCPS, but we do not pair that score with answer accuracy because generation used MinerU-on. Answer accuracy is 72.5% for Prod, 23.8% for MinerU, and 20.5% for PaddleOCR. In this three-parser check, the top-ranked parser also has the highest judged answer accuracy, while the nearly tied lower pair reverses. Because the same model generates and judges answers, we treat this only as a check of the top choice, not as validation of the full ranking.

Parser	Ans. acc.	EM	Answered	RCPS
Prod	**72.5**	**49.8**	87.9	0.583
MinerU (on)	23.8	14.5	39.1	0.137
PaddleOCR	20.5	11.5	45.9	0.140

Table 4: The RCPS top choice, Prod, also has the highest answer accuracy in this three-parser check; the near-tied lower pair reverses. Generation uses BGE-M3 top-five retrieval over 663 Q–A, with MinerU-on used for both RCPS and generation. RCPS averages the main three-retriever, three-cutoff grid. EM is normalised-span exact match; Answered excludes `NO ANSWER`. Ans. acc., EM, and Answered are percentages; RCPS uses a 0–1 scale. The same GPT-5.4 checkpoint generates and judges answers, so this checks only the top choice.

C Is the Absent Label Robust?

This appendix assesses the exact-span absent label with alternative matchers, a cross-family judge, and a parser-masked human check.

Parser I/O and the "absent" label. In our pipeline, each PDF page is rendered as an image, and a document parser maps that image to a linear Markdown transcription intended to represent body text in reading order, Markdown ATX (#-style) headings, and GitHub-flavoured Markdown tables. Layout coordinates, fonts, figure imagery, and page furniture are discarded unless they are transcribed as text. The documents include born-digital and scanned PDFs, multi-column layouts, merged table cells, stamps, seals, and text embedded in figures. Only the resulting Markdown is chunked, embedded, and retrieved. A reference answer receives the operational *absent* label defined in §3.2 when its normalised exact span has no match in the source-page transcription. The reference Markdown used for TextNED and the reference answer spans are derived from manually de-noised Qwen3-VL-30B output, which we treat as pseudo-ground truth.

Worked example: a corrupted table unit. On KoGov page `val_0155`, the pseudo-reference and parser output differ as follows:

Pseudo-reference answer	$A = 180\,\mathrm{m}^2$
MinerU-on output	`A = 180m`

MinerU-on omits the exponent. The shared normalisation removes formatting but does not treat m^2 and m as equivalent. Under the exact-span relevance rule, the complete reference answer is therefore absent from this parser output, and rechunking cannot reconstruct it.

To examine whether shared Qwen3 lineage among the reference Markdown, reference answers, and Prod explains the absent-rate gap, we apply five alternative presence matchers to all 663 Q–A. L0 uses raw substring matching. L1 applies Unicode Normalization Form KC (NFKC), case folding, whitespace normalisation, and Markdown normalisation. L2 also removes digit separators and in-content punctuation. L3 requires at least 90% order-free token recall, while L4 requires the longest common substring to cover at least 80% of the answer. L3 and L4 are alternative tolerant criteria rather than nested relaxations. If surface form accounts for most of the gap, the gap should narrow under at least one tolerant matcher.

Absent %	L0	L1	L2	L3	L4
Prod (2B)	24.1	20.2	19.6	24.1	16.9
30B teacher	22.8	18.9	16.6	19.2	16.3
2B base	30.0	25.2	24.7	27.0	21.1
MinerU-off	74.8	70.4	68.0	73.0	68.6
PaddleOCR	67.0	62.7	62.0	59.9	60.5
MinerU−Prod	+50.7	+50.2	+48.4	+48.9	+51.7
PaddleOCR−Prod	+42.8	+42.5	+42.4	+35.7	+43.6

Table 5: The OCR-parser absence gaps over Prod persist across five presence matchers (663 Q–A, no retriever; MinerU-off). L0 is raw matching; L1 is normalised exact matching; L2 removes additional punctuation; L3 uses token recall; and L4 uses longest-common-substring coverage. The two other Qwen3-derived parsers remain within 5.9 percentage points of Prod.

We next ask GPT-5.4, a different model family from the evaluated parsers, to classify each parser–answer pair among the L1-absent cases by whether its evidence remains recoverable for retrieval (Table 6). *Present* denotes a recoverable surface mismatch, *degraded* means that answer evidence remains in the parser output but is too corrupted to recover reliably, and *absent* denotes missing content. We combine *degraded* and *absent* as *retrieval-unusable* cases and apply the same procedure independently to each parser. For MinerU, this analysis uses MinerU-off. Because the Q–A were also generated by a GPT-family model, this judge is not independent of the Q–A generation process; the model-free matchers remain the primary evidence.

The judge marks 56% of Prod's L1-absent cases as surface artefacts, compared with about 16% for MinerU and PaddleOCR. The alternative judge therefore does not eliminate the MinerU–Prod gap: their retrieval-unusable rates differ by 50.4 percentage points. Comparable gaps appear under L1 exact-span matching (50.2 percentage points) and the L4 OCR-tolerant criterion (51.7 percentage points).

Human verification of absent labels. Two authors independently labelled a parser-masked, question-type-stratified sample of 100 L1-absent cases: 50 MinerU-on, 30 Prod, and 20 PaddleOCR. This study tests coverage labels, not the separate 100-pair Q–A quality sample. Agreement was $\kappa = 0.615$ and 81/100 before the remaining 19 cases

	present (artefact)	degraded	absent	retrieval-unusable
Prod	56%	2%	42%	**8.9%**
MinerU-off	16%	9%	75%	**59.3%**
PaddleOCR	16%	22%	63%	**52.8%**

Table 6: A cross-family judge preserves the OCR-parser gap after recoverable surface mismatches are separated. GPT-5.4 labels 1,017 parser–answer cases marked absent by L1 (134 Prod, 467 MinerU-off, and 416 PaddleOCR). Degraded and absent are retrieval-unusable; the first three columns are percentages within each parser's L1-absent set, whereas the final column is a percentage of all 663 Q–A.

were jointly adjudicated. After adjudication, 42/50 MinerU-on cases were retrieval-unusable (84.0%, Wilson 95% CI $[71.5, 91.7]$). The corresponding results were 12/30 for Prod (40.0%, $[24.6, 57.7]$) and 19/20 for PaddleOCR (95.0%, $[76.4, 99.1]$).

Of the 100 human-labelled cases, 93 could be matched to the automated-judge set: 43 MinerU cases, 30 Prod cases, and 20 PaddleOCR cases. Seven MinerU-on cases had no corresponding MinerU-off instance and were excluded. On the 93 matched cases, the judge's retrieval-unusable rates were 84.2% for MinerU, 84.1% for PaddleOCR, and 44.0% for Prod; pooled binary human–judge agreement was 84/93 (90.3%). The human labels therefore confirm the direction of the OCR-parser gap on this stratified sample. Different sampling fractions and output configurations, however, preclude a population-level replication claim.

Causes of absence. For Prod, exact-match absent counts are 23/165 (13.9%) for table-evidence answers, 43/201 (21.4%) for factoid answers, 63/290 (21.7%) for procedural answers, and 5/7 (71.4%) for figure-evidence answers. The figure estimate has only seven cases and should not be generalised. Its high point estimate is nevertheless consistent with figure imagery not being transcribed into retrievable text. We observe three recurring causes. First, parsers drop table cells before they reach Markdown: among table-evidence answers, absent rates are 87.9% for MinerU-off, 41.7% for MinerU-on, and 13.9% for Prod. Second, parsers skip text embedded in captions, stamps, seals, or figure labels. Third, they misrecognise numerals or units beyond the L2/L4 tolerance.

D Chunkers Rarely Split Exact Spans

Across eight chunkers, exact-span splitting reaches at most 2.3%, while the parser-defined absent rate remains 20.2%. This retriever-free diagnostic uses Prod output on 294 pages and all 663 Q–A; text matching labels each answer as covered, split, or absent, as defined in §3.2.

Chunker	covered	split	absent
md_h3	79.8%	0.0%	20.2%
md_h2	79.8%	0.0%	20.2%
md_h1	79.8%	0.0%	20.2%
parser_native	78.1%	1.7%	20.2%
fixed500	77.5%	2.3%	20.2%
fixed500_ov200	79.8%	0.0%	20.2%
fixed1000	79.3%	0.5%	20.2%
fixed1000_ov200	79.8%	0.0%	20.2%

Table 7: No tested chunker splits more than 2.3% of answers; the pre-chunking absent rate stays 20.2% (Prod, 294 pages, 663 Q–A). `parser_native` uses blank-line paragraph boundaries; md_hk uses Markdown heading depth k; fixed N[_ov$M$] uses N-character windows with optional M-character overlap.

E How RADP-DPO Builds Preferences (C4)

Common training setup. All variants are LoRA fine-tunes of Prod ($r = 8$, $\alpha = 32$) (Hu et al., 2022). Evaluation uses deterministic decoding (temperature 0; 1,536-token limit), while candidate generation for preference construction is stochastic. The 2,667-page training corpus and its 6,164 generated Q–A are page-disjoint from every evaluation frame. RADP-aux sweeps $\lambda \in \{0, 0.1, 0.3, 0.5\}$ for its contrastive loss. Unless stated otherwise, confidence intervals use 1,000 paired Q–A-level percentile-bootstrap resamples.

RADP-DPO samples alternative Prod outputs on the separate 2,667-page training corpus. For each candidate parse, the reward uses questions linked to that page, with other-page chunks as distractors; the corpus contains 6,164 Q–A in total. The reward is page-local BGE-M3 MRR averaged over $k \in \{1, 5, 10\}$, rather than the three-retriever RCPS used for evaluation. Candidate pairs with a sufficient reward gap become DPO preferences; multilingual-e5-large and Qwen3-Embedding-8B remain unseen by the reward.

For each training page, we sample K parses, chunk each output, and score its chunks against the page's questions plus distractors. When two candidates differ by at least 5 percentage points in page-local reward, the higher-scoring parse becomes the chosen example. R1 samples $K = 2$ parses with uniform distractors. R2 initialises from the R1 checkpoint and trains on a second preference round. R3 expands the pool to $K = 16$ by adding 14 new parses to the two R1 candidates. It also adds full-corpus hard negatives: chunks from other pages that are closest to the current queries.

Training objective. Let x denote the input page, and let y^+ and y^- denote its chosen and rejected parses. We define $\Delta_\theta = \log \pi_\theta(y^+ \mid x) - \log \pi_\theta(y^- \mid x)$ and $\Delta_{\text{ref}} = \log \pi_{\text{ref}}(y^+ \mid x) - \log \pi_{\text{ref}}(y^- \mid x)$. During training, π_θ enables the trainable LoRA adapters, whereas π_{ref} disables them:

$$\mathcal{L}_{\text{DPO}} = -\log \sigma(\beta[\Delta_\theta - \Delta_{\text{ref}}]). \qquad (2)$$

The SimPO control uses a length-normalised chosen–rejected log-probability margin ($\beta = 2.0$, target margin $\gamma = 1.0$) without a reference policy.

F Parser-Training Results Across Frames (C4)

Across frames, observed training differences are small or uncertain: the 73-page pilot misses its target, every DPO and SimPO interval in the 242-page KoGovDoc-RAG comparison crosses zero, and OHR-Bench compatibility estimates are about one Hit@5 point.

Frames. Before the pilot, success meant at least 5 RCPS points with a 95% CI lower bound above zero. The pilot held out 73 pages and 202 Q–A from 242 evidence-bearing pages; its best RADP-aux and RADP-DPO estimates missed the target and had intervals crossing zero. The later 242-page analysis pools the 169 development and 73 held-out pages, so it is not an independent holdout. All 663 Q–A have evidence on these 242 indexed pages; the 294-page selection index adds 52 Q–A-free distractor pages. Table 8 instead uses a 2,036-Q–

A, six-domain OHR-Bench compatibility subset. All candidates come from the separate 2,667-page training corpus, and all evaluated systems are LoRA fine-tunes of Prod.

On KoGovDoc-RAG, the three RADP-DPO milestones have parser-native Hit@5 estimates 1.96–2.11 percentage points above Prod. The paired-bootstrap proportion $P_{\text{boot}}[\Delta\text{Hit@5} > 0]$ is approximately 0.90 for each (Table 9); this descriptive proportion is not a p-value. Every two-sided interval crosses zero at $n = 663$. These estimates are exploratory because this reporting configuration was chosen after examining multiple chunker, retriever, and cutoff settings without a family-wise multiplicity correction. The bootstrap resamples Q–A pairs, not evidence pages. Table 8 omits R1 because no source-aligned cross-domain array is available for that checkpoint. The RADP-Distill comparison is also omitted because no aligned per-Q–A artifact is available for the same evaluation frame.

RADP-aux does not meet the prespecified criterion. On the held-out 73-page, 202-Q–A fold, $\lambda = 0.1$ gives the largest estimate relative to the $\lambda = 0$ fine-tuning control: +1.13 RCPS percentage points with md-h3 and +2.31 with parser-native chunking. Estimates decline at larger values of λ, and every paired confidence interval relative to $\lambda = 0$ includes zero.

SimPO has negative point estimates. Relative to Prod on the 242-page fold, its three-retriever mean Hit@5 difference is −0.85 percentage points with md-h3 and −0.70 with parser-native chunking; both confidence intervals cross zero. These results do not isolate whether reference removal, length normalisation, or another optimisation difference causes the change.

Unavailable fidelity control. RADP-Distill ranks candidates by edit distance rather than retrieval, but an aligned per-Q–A artifact is unavailable. We therefore omit its cross-retriever and evidence-type breakdowns and make no quantitative comparison between training objectives.

G Do Training Point Estimates Track Fidelity or Boundaries? (C4)

The available measurements do not identify a training mechanism: DPO checkpoints have lower TextNED than Prod, but boundary evidence is incomplete and TextNED does not order their retrieval results. Table 10 reports Boundary Clarity (BC), normalised edit distance to reference Markdown (TextNED), and chunking shape on the 242-page corpus. TextNED is character-level Levenshtein distance divided by the longer string length.

Variant	len	c/pg	clen	BC↑	TextNED↓
Prod (baseline)	1385	4.82	274	0.630	0.240
aux λ=0.0	799	2.96	245	0.553	0.626
aux λ=0.1	1380	4.05	321	0.652	0.423
aux λ=0.3	1930	6.73	272	0.470	0.332
aux λ=0.5	2035	5.82	327	0.556	0.318
DPO-R1	1606	4.97	311	0.646	0.167
DPO-R2	1610	4.83	321	0.647	**0.163**
DPO-R3	1514	4.88	300	—	0.185
SimPO	1703	5.31	305	0.601	0.186

Table 10: DPO checkpoints have lower TextNED than Prod on the 242-page corpus, but the metric does not reproduce their retrieval order. Len is mean parsed characters per page, c/pg is mean chunks per page, and clen is mean characters per chunk. BC was not computed for R3.

DPO checkpoints have lower TextNED. The three RADP-DPO checkpoints have TextNED values of 0.163–0.185, compared with Prod's 0.240; these are 23–32% lower relative to the pseudo-reference. RADP-aux instead has values of 0.318–0.626. All three DPO checkpoints also have positive Hit@5 point estimates in Table 9, but TextNED does not order them: R2 has the lowest TextNED and smallest DPO Hit@5 difference, whereas R3 has the highest TextNED and largest difference. This co-occurrence is consistent with text fidelity as one possible explanation, but the observational statistics do not identify a causal mechanism.

Available structure evidence is incomplete. R1 and R2 have 4.97 and 4.83 chunks per page, close to Prod's 4.82, while their mean chunk lengths increase from Prod's 274 to 311 and 321 characters. Their BC values are 0.646 and 0.647, compared with Prod's 0.630. R3 has comparable chunking shape but no reproduced BC value. Missing R3 BC and un-

Δ vs Prod (pp)	Hit@1	Hit@5	Hit@10	MRR@10	nDCG@5
RADP-DPO-R2	+0.59	+0.95	+0.90	+0.78	+0.82
RADP-DPO-R3	+1.46	+1.15	+0.90	+1.30	+1.28

Table 8: R2 and R3 have small positive deltas on the audited OHR-Bench compatibility subset, but this is not a full v2 rerun. The subset contains 2,036 source-aligned Q–A across six domains and uses a three-retriever average with 1,000 Q–A-level paired-bootstrap resamples. Deltas are percentage points; Hit@5 intervals are R2 $[+0.33, +1.54]$ and R3 $[+0.31, +2.05]$.

Variant (vs Prod = 0.6757)	Hit@5	ΔHit@5 [95% CI]	$P_{\text{boot}}[\Delta\text{Hit@5}>0]$	ΔRCPS (pp)
RADP-DPO-R3	0.6968	$+2.11$ $[-0.90, +5.13]$	0.91	+1.72
RADP-DPO-R1	0.6963	$+2.06$ $[-0.96, +5.13]$	0.91	+0.57
RADP-DPO-R2	0.6953	$+1.96$ $[-1.06, +5.03]$	0.90	+0.47
RADP-SimPO (ctrl)	0.6687	-0.70 $[-3.77, +2.31]$	0.32	−1.56

Table 9: All KoGovDoc-RAG DPO point estimates are positive, but every two-sided interval crosses zero; the SimPO point estimate is negative. The analysis indexes 242 pages for 663 Q–A and uses parser-native chunking with 10,000 Q–A-level paired-bootstrap resamples. Hit@5 averages three retrievers; Prod is 0.6757. $P_{\text{boot}}[\Delta\text{Hit@5} > 0]$ is a bootstrap proportion, not a p-value. Across three independent R1-recipe seeds, the standard deviation of mean ΔHit@5 is 0.90 percentage points.

certainty estimates prevent a systematic conclusion about boundary change.

Interpretation. One possible explanation for RADP-aux's smaller point estimates is that its hidden-state objective can affect retrieval only indirectly through generated Markdown. Previously computed OHR TextNED and RADP-Distill breakdowns use artifacts that are not aligned with the audited subset, so we exclude them from the mechanism analysis.

H Dataset Composition and Release Status

Dataset composition. KoGovDoc-RAG contains 229 Korean government pages and 65 arXiv pages. These sources contribute 527 and 136 verbatim-answerable Q–A, respectively. The separate 2,667-page parser-training set and these 294 evaluation pages are page-disjoint. Prod exceeds its 2B base by 4.9 Hit@1 percentage points, compared with the 42.6 percentage-point parser-selection difference. This comparison provides scale only; it is not a formal bound on fine-tuning effects.

Reproducibility. The repository currently contains the frozen 663-Q–A probe, MinerU-on page outputs, retrieval and diagnostic code, per-Q–A arrays for the audited training comparisons, and aggregate artifacts for the main parser and chunker grids. A deterministic script also reproduces the 2,036-Q–A OHR compatibility mask and paired estimates from the stored arrays. The public tree does not yet contain the MinerU-off page outputs, final per-case human adjudication labels, an aligned RADP-Distill artifact, complete executed-checkpoint configurations and checkpoints, or per-Q–A arrays for the complete 294-page grid. A full OHR-Bench v2 rerun and clean-checkout command verification also remain outstanding. We do not present these items as released or reproduced.